History of Science Lecture Notes

geschreven door

gg2000

HC1 – Introduction

Why this course?

Out of comfort zone! Reflection on what we are, why we are (where we came from), where are we heading (and do we want to take another turn)?

The most important reason [to examine and build ICT professionalism] stems from the extent to which the increasing pervasiveness of ICT has the potential to harm our economy and society. The extent to which ICT is embedded in our lives is inevitably growing. If we fail to take steps to mature the ICT profession, it is likely that the risks to society from ICT will grow to unacceptable levels.

Examples:

1. Influence on our use of language or habits, use of metaphor
2. Augmented reality used for faking "news", counting seconds in health care
3. Relationships (the movie Her)

Historical:

- Distance in time makes it easier to recognize several actors and interests
- Distance in time makes choices from the past more visible
- Historical failures won't help to prevent making new ones, but it does make hidden assumptions more easily recognizable.

Sociological (for example humor):

- By describing the information sciences as a (sub)culture in its' own right
- With specific habits and "rituals"
- Emphasis on rise of a profession

History of Science

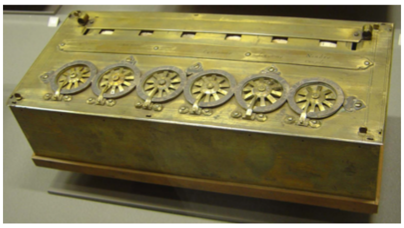
Prehistory

Roughly: Campbell-Kelly chapters 1-2. Ideals made machines into a success!

Machines and Technology

We take it for granted, because machines and technology are self-evident

- The chess-playing turk, you can play against a puppet, mechanical thinking was presented, but it was actually a dwarf in the machine
- Trust in technology is not self-evident
- Calculating devices

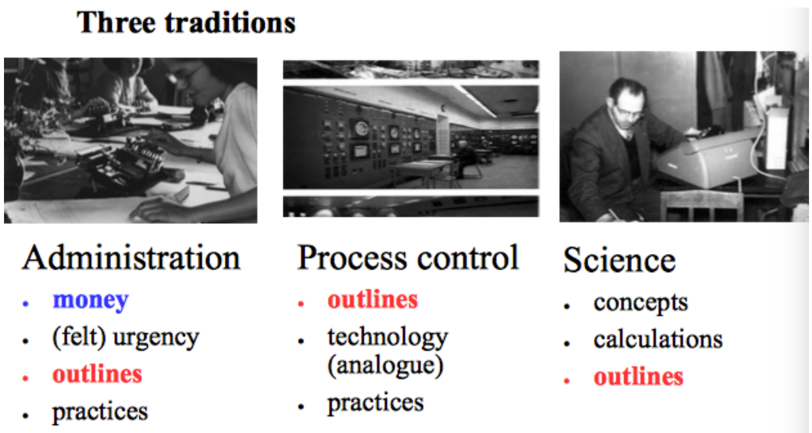


Charles Babbage (1791-1871)
On the economy of machines and manufactures.

Frederik Taylor (1856-1915): shop management (1903), the principles of scientific management (1911), Ford (1913). Afterwards, Henri Fayol (1841-1925), administration industrielle et generale (1916).

Trust in numbers and efficiency
- US census (1890)
  - Trust in Hollerith machines
- The value of statistics
  - Diagram of the causes of mortality in the army in the east
  - European continent was politically more "leftish" than US/UK

Three traditions
1. Administration
   a. Punched card typing, calculators, typewriters
   b. Sorting machines, counting machines, tabulating machines
   c. 1923: efficiency movement
   d. Not necessarily all very high tech, but following a business outline
2. Process control
   a. Outlines of analogue to the process
   b. Outline of process
3. Science and engineering
   a. Computers (PEOPLE!) active in:
      i. Weather (storm) predictions
      ii. (Hydro)mechanical calculations
      iii. Aeronautics
      iv. Econometry (Jan Tinbergen)
      v. Telephone cables
      vi. Military applications (tables for aiming large guns; Manhattan project)



<u>Administration</u>: Automation. You can find working machines here.
<u>Process contro</u>l: working computers (analogue, reacting to processes).
<u>Science</u>: people build machines just because they could, to make things easier.

HC2 – Creating Computers

Assignments HC1

The profession of the "computer scientist" (Babbage). Babbage: De Prony, Hollerith (names!) are not always important, but they do provide facts. More names in chapters 3-4! Re-invention (Babbage).
Three traditions: changing habits (chasing antelopes across the Serengeti vs producing eligible handwriting, or being able to read it). We read less handwriting, but this is only the changing of society.
The word computer is difficult to pinpoint: AC = automatic calculator.

Recap

Today we will focus on the ideals that made people bother. Why after WW II people started putting faith in machines that evidently did not work. And how they got them working!
ATTENTION: This means a shift with respect to C-K.

Z1 (1938): Zuse is inventor of the computer (het built the Z1, Z2 etc.), you can't call it a computer, but automatic calculators!

Cold War Science (background)

End of WWII; **Manhattan Project** = building machines, get scientists to work together, trying to focus on building a machine that can calculate, building the atomic bomb and other mechanical things.

Code Breaking (Bletchly Park): Alan Turing, enigma.

Vannevar Bush (*1890-1974*): Manhattan Project (issued by the government). In *1945*: the atlantic monthly; "so many people (pc people and engineers) have been working against the war, the government should continue funding these people". Two reasons:

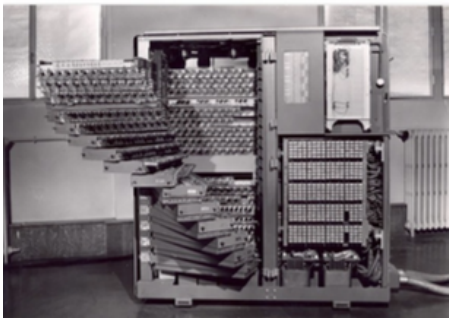
1. he believes that science can prevent war
2. he is pragmatic and he knows that if all those people would become jobless, it would be bad

After WWII, the need for scientific calculations exploded. **NASA** is best known for the trip to the moon, but rocket technology was only available with atomic power (Shell).

In (continental) Europe of the late *1940s* as well: rebuilding the nation (**Marshall plan**). Mix of fatalism (because people knew that the war had been won, but the cold war had been started at the same time) and optimism (because the war was over!!).

Mathematisch Centrum (*1946*): director is ???. Van Dansig and Van Wijngaarden fully believed in science as the way to rebuild the Netherlands.

Hans Fredenthal (*1905-1990*): ondergedoken in oorlog. "Science will help our nation, society could improve!". May *1949*: colloquium modern automatic calculators (with Van Wijngaarden and Freudenthal).



1950: Time Magazine "Can man build a superman? Harvard Mark III soon to be delivered to the Navy".

Jean Jacques Servan-Schreber (1924-2006): *Le Défis Américain* (*1968*) = the American challenge: producing our own European mathematical machine. All AC's weren't working good yet, so why shouldn't Europeans build their own computer? → Gamma1.

Dinosaurs (what was the first computer?)

Manchester Baby, and another dinosaur weren't working very well. They operated, but they needed a lot of human intervening.

Amsterdam

Aad van Wijngaarden (1916-1987), from *1947* onwards director of the "computing department" of the Mathematisch Centrum.

Jan van der Corput (1890-1975), from *1950* onwards general director of the Mathematisch Centrum. *1952:* *"Nobody doubts the innocence of automatic calculators, but, like any piece of technology, it is constructed for a specific purpose, and once in existence it can be used for good and for evil. You can't wish them away. Although these machines were devised for destruction, it is now our duty to put them to good use. They might lead to the loss of labour, with all the horrible social consequences that go with that. But they might also help to prevent disaster in the future."*. **He wanted to rebuild the Dutch Nation and prevent WWIII**.


ARRA, het technisch rekenwonder

jaar	R	O	MC	ARRA	ARRA II	ARMAC	X1
1946							
1947							
1948							
1949	*64*	9					
1950	*114*	52	27	**2**			
1951	*166*	59	21	**1**			
1952	*206*	48	17	**1**			
1953	*249*	52	13				
1954	*297*	59	8		**8**		
1955	*340*	53	13		**20**		
1956	*394*	60	9		**5**	**13**	
1957	*467*	73	9			**38**	
1958	*514*	57	5			**28**	
1959	*557*	55	6			**17**	**2**
1960	*605*	69	7			**1**	**42**
1961	*725*	122	11				**122**
1962	*852*	179*					**179**

ARRA (1953): press button and machine goes silent. It never worked again.

"Bluffing their way into the computer age" - Gerard Alberts.

ARRA II (1954): soon at the disposal of the Dutch waterworks.

Vienna (1955), **ARMAC** (1956), Electrologica (1956): **X1**.

Delft

Willem van der Poel (1926), school of prof. dr. N.G. de Bruijn (optics and fiber technology). *1950:* **ARCO** (Automatic Relais Calculator for Optical problems), later called: *Testudo* Relais. Programs hard-wired, but interchangeable.

Ten behoeve van het telefoonverkeer (PTT) *1951*: **ZERO** (Zeer Eenvoudig Reken Orgaan),

1953: **PTERA** (PTT Electronische Reken Automaat).
1955: Van der Poel designed **ZEBRA** (Zeer Eenvoudige Binair Reken Apparaat).
In production 1957: Standard Telephone and Cable Company (**STANTEC**).

Eindhoven

PETER (Philips Elementair Tweetallig Elektronisch Rekenapparaat). Built by Wim Nijenhuis with the purpose: Trying to test whether the audio equipment was performing well. *1953-1956:* Testing (agreement with IBM). Gentleman's agreement (Phillips). Peter was used for **Philips NATLAB**: it was used for calculating. Afterwards they were called:

- **PASCAL** (Philips Akelig Snelle Calculator).
- **STEVIN** (Snel Tel- En Vermenigvuldigings Instrument).

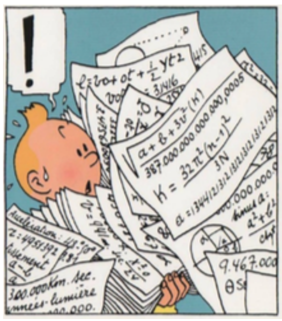
All these people later got involved with one another, either in the Colloquium "Moderne Rekenmachines" (since 1952) or in the development of ALGOL 60. All of the machines above were built for very specific purposes, for optical calculations, telephone calculations or acoustic calculations. They weren't built for memory or text processing, but for calculations.

1984


European tradition of dystopian literature. Aldous Huxley, *Brave New World* (1932). Popular expression: *Metropolis* (1927).

1984: George Orwell, *1984*: dictatorship novel.
- System Thinking: organized in a way that has to follow its path. That makes people relate to it.
- "big brother is watching you"
- Themes of the book:
    - Totalitarianism, nationalism, surveillance, censorship
- Who controls the past, who controls the future?
- Who controls the present, who controls the past?

"The Public"



All Cold War projects were costly and had to be sold to the general public.

People were scared of atomic power, so how to sell the idea of having an atomic nuclear reactor near your city? → Documentaries, movies etc. on nuclear power, space travel and the blessings of science in general.

Historically, the rise of computing technology as part of modernity can be pinpointed to the 50s. Hergé, *Les aventures de Tintin: Objectif Lune* (1953).

1956: Forbidden Planet, Robbie the Robot
1957: Desk Set

In 1956/57 it became common!

Golden age of Science Fiction (1950-1965):

1. Isaac Asimov (1920-1992):
   - *I, Robot* (1950)
   - *Foundation* (1951)
2. Robert Heinlein (1912-1994):
   - *Starship troopers* (1959)
   - *Stranger in a strange land* (1961)
3. Arthur Clarke (1917-2008):
   - *Interplanetary flight* (1950)
   - *Childhood's end* (1953)
4. Philip K. Dick (1928-1982):
   - *Solar lottery* (1955)
   - *The minority report* (1956)
   - *Do androids dream of electric sheep?* (1968)
5. Pierre Boulle (1912-1994):
   - *La planète des singes* (1963)

“will the coming of ACs change our lives?” They couldn’t imagine how. Public attention on radio, newspaper and television. Public perception of computing would change even more. AC became THE icon of progress in the late 1950s.

HC3 – The sound of software

Recap last week

The word software didn’t really exist yet (anachronism).

- Dinosaurs: From distrust in machines to trust in dinosaurs in different traditions. Trust in machines that did not work: tubes or relais.
- Mathematical Centre (1946): ideals in Cold War Europe or US
- Cold War: ambivalence towards US Défis Americain (1967)
- Most of the computers were made for calculating

EXAM: Be specific, but don’t renounce your own future profession. The question about the first computer requires you to define a computer. That can, of course, be in technical terms, but also in terms of use. That depends on your point of view!

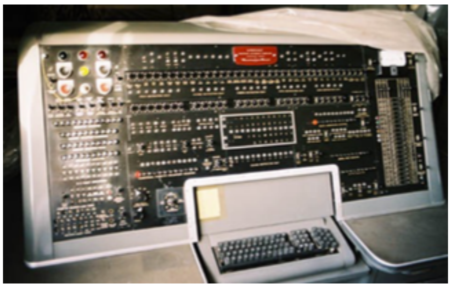
- Binary grading system (either it’s right or wrong), so show them what you know!

Computing Sounds

The idea of programming a computer emerged in the war (1946 w.r.t. ENIAC). Programming was used as a verb for the first time in 1948. It meant sticking plugs in a board, so it was a hard job. People started **appropriating** (trying to see what it can and can't do, get a feeling of the limitations). By listening to the machine you could see if it was working right.

EDSAC: Maurice Wilkes, Cambridge, 1949. Christopher Strachey sent a punched card tape suggesting to execute the code on EDSAC: humming "God save the queen".

Wilkes, Wheeler and Gill (1951): first handbook about programming (prof: "not a verb"). Old machines made noise, which helped with checking how the machine did.



- **ARRA** (1952) relais.
- **ARRA II** (1954) transistors: with speaker
- **Univac**: "stall speaker" button. The speaker was on.
- **Z22** (1957)
- **SAGE**
- Soviet Union: **BESM 6** (1963)

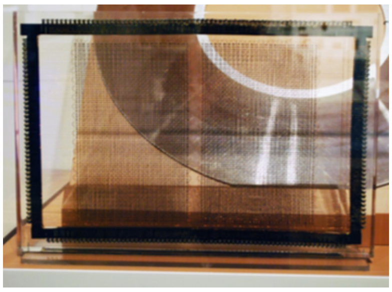
Sounds were used for:
1. Announcing the program had terminated
2. Auditive monitoring ("it works!")
3. Specific navigation in a program (mathematicians knew where the program was at)
4. Part of the instruction ("program is now here")

External control on execution.

Computing and use of metaphors

Use of metaphors is a way of dealing with new technology; coming up with names:
- Plugging, assembling
- Pseudo coding
- Assembling; assembly routine
- Autocoding, autocoding systems

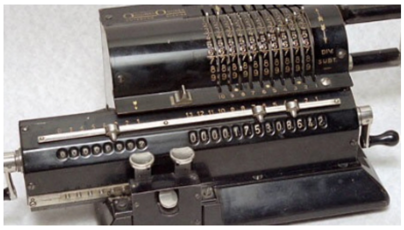
1960: programming language compiler became a translator. Reassuring names for new things.



Memory

Using cognitive psychology for naming things helps. The computers start defining the way we speak. Bolter (1984): "Defining technology". Memory was very useful for work processing, but computers weren't used for that. They were used for calculations. Until IBM came in.

IBM and (Dutch and other European) dwarves



Odhner

St. Petersburg (1873): **Odhner** (European IBM). After 1917 clones all over Europe.

Sweden: Åtvidaberg Industrie 1920, producer of successful line of calculating machines: **Facit**. 1970: bankrupt, sold to Electrolux.

Germany: Konrad Zuse (1910-1995) begins production of **Z4** in 1949 and **Siemens AG** in 1967. Siemens discontinued this line.

Vienna: Never was one, because Philips had the gentlemen's agreement with Siemens.

France:

- **Bull** (1930 office machinery. It was pretty convincing dwarf).
- Giant machines working and meant for business purposes (multipurpose). The French government supported the building of these machines:
    - **Gamma 3** (1952),
    - **Gamma 60** (1958); costed a lot of money to produce and the selling's were not as well as they've hoped → agreement to put in some money into the company and sold most of it to General Electric.
- 1962: **GE**
- 1967: Josh Schreiber
- 1968: **Honeywell**

Electrologica (1956)

- **Electrologica X1** (1959) produced in Amsterdam. Computer was moved to Rijswijk, Amsterdam
- **Electrologica X8** (1964)

They were usually for calculating, not for administration.

1966: The gentlemen's agreement came to an end and it was bought up by Philips. It was government funded.

1955, Denmark: Regnecentralen. Stockholders were also clients! Have them put a share in the company, and in that way they bought corporate loyalty.

1968: International Computers Limited; reasonable successful selling IBM compatibles. Merger of three older computer builders. 2002: Fujitsu.

In General

- Too late taking part in new developments
- Other economic circumstances:
    - Rebuilding the economy after the war
    - Small markets, limited advertising budgets
- Labour costs relatively high; therefor the costs for R&D relatively high as well
- Pré: anti US (thereby anti IBM) attitude

IBM was dominating in the market of office equipment, also in Europe. In 1960: 60% machines were IBM and 90% was non-European.

1959: **IBM 1401**.
Nickname = Big Blue, because it was blue but also because they were thought of as the Big Brother that constantly was trying to hurt the middle market share (so it was a negative reference). Sales Strategy: FUD (Fear, Uncertainty and Doubt).

1963: IBM System/360. Operating calculators.

The dwarves were ended, it's not a bad or a good thing. The US battled us with their huge amounts of money and strategies.

HC4 – Computing Crisis

Computer People who died

Robert Taylor (1932-2017).

Chriet Titulaer (1943-2017):
- director of the dutch "Television Academy"
- Promoting tech and science.

Saskia Stuiveling (1945-2017):
- President of Dutch Court of audit (algemene rekenkamer)
- Transparency in spending of government money and NGO budgets
- Open data
- Promoting the use of computing

Assignment 3

Big Blue: indeed, IBM produced big and blue machines, but "big" referred to the company itself, and to the dictator in Orwell's 1984. Metaphor question: language became a metaphor for code, not the other way around! Note the exam on blackboard.

Real time computing

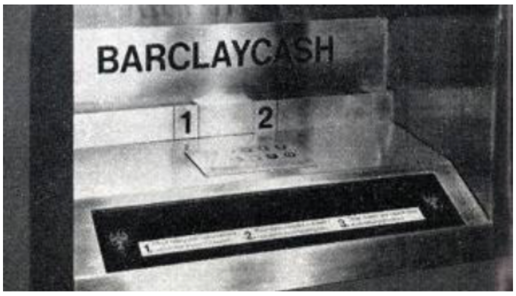
Computers were not designed to give a reaction in real time, but to process.

1944: Jay Forrester from NIT asked for war effort, assigned to the Special Devices Division. Goal: universal flight trainer.

SAGE

IBM, Burroughs, Bell labs delivering the machines learning about core memories, printed circuits, mass storage and programming (1800 programmer years). They were scared for Russian bombers, so they wanted a vision of the American airspace and you need real time computing for that. Operational 1963, cost $ 8 billion meanwhile: missile technology operational until 1980.

Whirlwind/SAGE: most important contribution to civilian computing through IBM and other contributing firms (SABRE, ATM, UPC). No longer some secret defense system, but the results of computing becoming available to the larger populace.



ATMs and Product Codes

Cash Dispenser (1967) Barclays, London: the robot cashier (ATM). Ideal of the “cashless society” (diners club). VISA (1974) continuing “real time”.

Universal Product Code: 1970-1971, formally adopted: 1973.
1974: IBM and NCR start producing scanners.

- Political feat (illustrating trust in real time computing)
- Effect on society: it became more difficult to enter the food market (you had to have the code and have the machines to go into the food business, so it put the whole food market upside down)

European Article Number Code (1973-1974), 12 countries in Europe. Meanwhile known as IPC (International Product Code).

Barclays and Burroughs

1965: looking ahead to D-Day (February 15,1971) = Decimal Day, the first day that the British would go decimal. The banks realized that they had to adapt their software.

The machine that Barclays wanted was an accounting machine and it had a connection to the mainframe that would do the main processing. These mainframes are TC500 and B8500 combi. They had a problem with the amount of time, so they downgraded. If the number is less, the computer must be less good too.

It lasted until 1971 until they decided that they couldn’t finish it, they just wanted a month of extra time. Real time was great, but on time would work as well. So there were different levels of real time introduced to the public.

SUMMARY: In Europe the need for real time computing was less felt than in the US. You didn't REALLY need it in business, but it COULD go well with banking. So change was not necessary. In the US, that process went much more quickly, because it was more necessary there.

Agenda's

What did people want to achieve?
- Selling machines
- Academic discipline (computer science)
- Thinking machines
- Telephone connections (Telstar AT&T, 1961)

Even within the academic world there were different **agenda's**:
- **Cybernetics** (Norbert Wiener, Claude Shannon etc.)
    - It's not an important part of the US perspective, but it is from the EU
    - Computational tech came from cybernetics
    - Trying to understand human thinking and behavior (playing chess, Turing)
- **Logic**
    - Constructing a language from a logical point of view on an AC
- **Sharing information**
- **Ordening / sorting / processing**
- **Calculations**

Programming

Wilkes, Wheeler and Gill (1951): first programming manual: "preparation of programs", "plugging", "assembling".

What were the alternatives? Within Whirlwind people were thinking of setting up the program in a more efficient way: "autocoding". **FORTRAN** (1957), **COBOL** (1959): Program started from "flowchart" and then they selected programmers to work on different parts of the flowchart and producing a whole program.

Programming language **ALGOL 60** in Europe**:**
- Language was seen as elegant
- Universal (it was an IDEAL: both machine and language were seen as universal)
- Satisfies European sense of clarity and order
- They were thinking from a mathematical point of view (except for France)
- ALGOL 60/68 was Europe's **frustration**. What was the problem that it was trying to solve? Making existing programs run on a new machine (you had to rewrite your whole program every time you got a new machine)
        - Writing a compiler would help → Ease of programming

Or from a US point of view:
- Language was seen as academic
- Not flexible (better in theory than in practice)
- Difficult to learn (too mathematical)

Who had interests in programming (agenda's)?
- ACM
- IEEE (= Electrical and Electro technical Engineers) → ACs
- IBM → what language to choose
- IFIP
- GAMM → ALGOL loving Germans
- SHARE (IBM)
- USE (Univac)
- DUO (Datatron)

The ALGOL conspiracy: dream of a universal language. Darmstadt, Dresden, München, Zürich, Mainz.
- IFIP: 1962, working group 2.1 = to start defining ALGOL as a language
- ACM: standard for publication of scientific algorithms
- SHARE: large community of IBM computers, ALGOL was too overly academic according to them
- IBM: the people who bought the IBMs didn't like the ALGOL language

Grace Hopper, ca. 1960: programming (she called programming "educating a computer"). She did a lot of COBOL teaching and was great at programming.

Rise of a software-industry
Beginning 1960s: software alone did not suffice to uphold a business → a range of computer services:
- Maintenance
- Machine building / tuning
- Batch processing

Term of "**software**" was vague in 60s: Code produced by one of the companies' programmers was rarely called "software" (it was called code or program). An external company testing a program or even just documenting was viewed as a "software project".

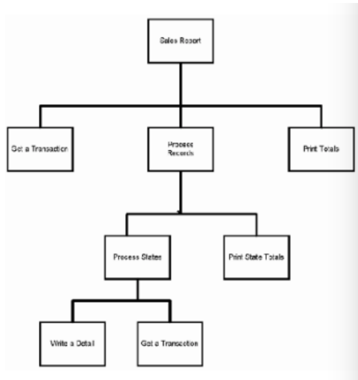
IBM 1968: "**unbundling**". From that moment onwards, software became both soft (not that attached to the machine) and ware (it had the common commodity in the 60s). Academics in Europa had some trouble with the idea that software was not something that was free but you had to pay for it.

The software crisis
Problems with "growing" amount of code SDC, the "university of programmers". This was a growing problem in the US. 1959: 700 programmers, 1400 support staff > 1.000.000 instructions (see picture p. 189).

IBM's OS/360 drama:
- The crisis was not really there in Europe.
- NATO?
- Banks?
- Companies had no crisis.

- Scientists did have a crisis.

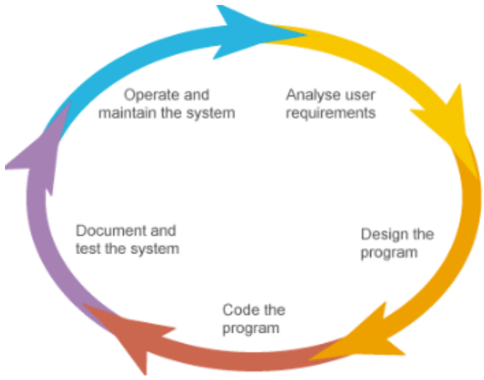
Everyone was calling themselves a programmer and made money with it. **Software engineering techniques** were needed to solve the crisis.



Garmisch-Partenkirchen October 1968, Ways out suggested:

- Universal programming languages? (ALGOL)
- Structured programming (programming on paper)
- Style: System development (methodology) and Project management.
- Economical: Finding more programmers.

Agenda was set by academics: software engineering primarily theoretical. Programmer became a profession, also in the economic sense. Informatics a science, software engineering being an important building block (if not foundation).



1972, Fred Brooks: Structured design. Use formal methods! **Structured programming**, proving program correctness (Turing, Dijkstra).
1967, Dijkstra: “The end of a vocation”.

Development model: software as a management problem (really American model).

HC5 – Dream machines or modes of computing

Recap

1976: Cold war: ambivalence towards US (Défis Americain). Information science was there before computer science could evolve or exist. For the first time producing machines that could actually work.

1944-1963: Real time computing

- flight simulator (didn’t succeed)
- civilian spin-off (product code and ATM)

Agenda’s (what people wanted to achieve):

- Selling machines
- Academic discipline
- Thinking machines

Software crisis: agenda was set by academics (software engineering primarily theoretical). Programmer became a profession and informatics became a science.

Computers were important in information science (automation).

Assignment 4

FORTRAN & COBOL: both solving the same problem!
ALGOL: not ONLY European, the frustration was only European
Software question: the origin of the phrase "software" begs for more than just its first use. First use can be a start (not necessarily), but changing meaning and use (and reasons behind that) is what I expect to read about.
Compare: origin of the word "computer"

Silicon Valley and European Valleys

Silicon Valley

US: computer part of the democratic movement, *time sharing* facilities. Meanwhile: developments in (micro-)electronics.

1968-1972: Whole Earth Catalog (Stewart Brand). This is the first time people actually saw a picture of earth from out of space. Nobody realized that it was computing that made the launch of this rocket possible.

Advertising in 1970s: a new audience. 1980: movies like Wargames and Back to the Future.

European Valleys

In Europe, mostly academics and high tech companies were interested in time sharing facilities.

Utopian city planning was mainly in the '60, but only these two survived:

- Sophia-Antipolis, France, 1969. One of the few that were successful, because people were fine during the week but wanted to go back to Nice in the weekends.
- Twente Polytechnic, Enschede, 1964. A lot of buildings, roads, all rectangular and it kind of worked in the '90s. In 1964 it did work as a campus for students, but the big companies were not as enthusiastic as they had hoped.
- People didn't want to live there because there were no bars, no cinemas etc.

But there was no interest for computers within the rising youth movement. This is because of the hippies in Europe, and in the Netherlands specifically they were not that politically oriented as the US. They wanted to check on the government because they didn't trust them, so they were a bit interested in computers.

Appropriating the machine: dream machines? [NL]

1975, **Altair 8800**: remembering numbers, the people loved these kind of machines to make programs on.

Appropriating = Making the computers your own, using it for your own agenda, not using it for their intentions but appropriating in the way YOU liked. 1975: Homebrew Computer Club.

1977, **Hobby Computer Club** (HCC) in the Netherlands:

- activities for the members
- exchanging knowledge and software
- programs in the newsletter
- radio-broadcasting programs with NOS
- no political agenda
    - like the hippies *(I think he said that)*
- identity, though!
    - Seen as amateurs, but with knowledge, passion and dreams

PET Benelux Exchange (1979), Philips Thuis Computergebruikers (1981) and Micro Computerclub Nederland (1984): In the course of the eighties they changed, becoming customer organizations tied to the machines the people were favorably using. HCC and PTC mainly: guarding the interests of users.

1982: **Commodore 64**, writing small programs. **Sinclair ZX Spectrum**.
1980s: **Squatter Movement** = protesting against the rising prices of properties in big cities, they would occupy houses and let people live there in a kind of communal way.

Chaos Computer Club: striving for overthrowing the government. They were fighting for some freedom in German/Netherlands virtually. They exposed weaknesses in certain systems and helped preventing real disasters.

- Schemes for modems (DM 10,-)
- Reviewing
- Software-ideas and plans
- Political activism

1. Lack of political activism in 60s and 70s
2. Rise of computer science in universities
3. There WAS continuity in NL (cf Campbell-Kelly) - NOT IN United States!!
4. And there was no academic setting for Information Science (although that didn't exist in an academic setting in the Netherlands; it did in the US, France and UK though)

These businesses started using computers early on for doing simple administration calculations. The computers were not self-evidently Dutch or even European.

- Central Statistical Agency
  - (CBS, Hollerith since 1934 - Electrologica 1960 - IBM 650 was considered)
- Dutch State Mines company
  - (Bull, 1958): *Calculating to get into the mines*
- Dutch Railroad company
  - (Bull, 1959): *To calculate the materials that were transported*
- National Tax Office
  - (IBM, 1959): *Calculating taxes*
- Insurance companies (Electrologica, 1960)
- Rijkspostspaarbank (after failure 1922 - IBM, 1960)

The people responsible for automation are the people who are seen as computer scientists today. The leaders of these (above) groups of people started a study group for automating administration:

- 1958, SSAA: Stichting Studiekring Administratieve Automatisering
- Journal: *Informatie*
- The firsts who provided information, courses and seminars
- H.J. van der Schroef (1900-1974)

Max Euwe (1901-1981):

- UNIVAC ambassador, advisor in automation, high chess player
- 1964: professor in Automatic Dataprocessing (Rotterdam Handelschool)

1979, the “more serious” users: VisiCalc (first spreadsheet for example).

S.S.A.A. H.C.C.

Funded by the Dutch government: Teleac television courses about microprocessors and programming to get people interested and knowledgeable about computers.

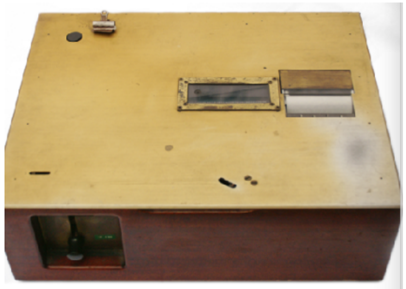
Computers in education

The computer became powerful because of the dreams so many people had and the potential so many people saw in it. People related to it. This was all before the computer was actually THERE in the classroom:

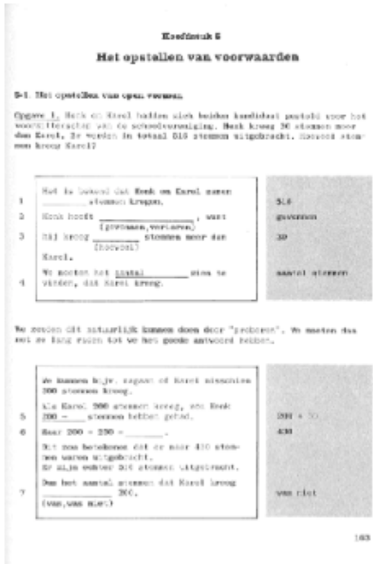
- Computers in the curriculum
- Computers in thinking about education (psychology)
- Computers in the educational process
  - computers in the classroom
  - computers (or computing influence) in textbooks

What was taught about computers in mathematics education:

- Binary arithmetic
- Flow charts
- Programming language École, BASIC
    - The language that you were teaching the students; programming IS mathematics
- Limitations and capabilities of the computers
- Book keeping courses



GOAL of computer in education: people give the answers that you expect from them and you want them to think independently. Solving equations doesn't necessarily involve thinking. Learning abilities: finding out how it works, new requirements, trying to meet the needs of all individual pupils and meeting the need of teachers.


Hoofdstuk 5

Het opstellen van voorwaarden

1954, Skinner: computer projects a question in the window and you can write down your answer and if it was right you could continue to the next question. This experience came from the military to quickly drill the engineering students in the military.

Programmed Instructions books, late 50s in US and spread across the globe in a few years.

- Influenced by computing technology on several levels
- Positive influence on general trust in computing technology

1970s: computing facilities were still rare.
1980s: computing facilities in classrooms.

HC6 – Personal Computers

Recap

Assignment 5

- Discuss in what ways Ted Nelson's Computer Lib movement influenced computer culture in the US.
- Information Science as an academic discipline was rather late in the Netherlands, although it institutionalized quite early. Explain this. Don't forget to compare the Dutch situation with France and the US!!
- Differences Homebrew Computer Club (bill gates advocated paying for software successfully, so it became a crime to copy software from within the HCC) and Hobby Computer Club (tried to use the fact that copying was happening as a way to get prices for software down): **difference was paying for software!**

Recap:

- From distrust in machines to trust in dinosaurs in different traditions. Trust in machines that did not work: tubes or relais
- Cold War: ambivalence of EU towards US. Défis Americain (1967). European dwarves.

- Playing with the automatic calculators, appropriating the machine started with the arrival of new machines, role of sound (transistors)
- IBM sales tactic: FUD (Fear, Uncertainty and Doubt)
- 1944-1963: Real time computing
- Civilian spin-off: ATM and product codes
- Agenda's (selling machines, academic discipline and thinking machines)
- Programming, program library (strips of code hanging down) and unbundling IBM (1968), which affected the rise of software
- Software crisis, programming became a profession and informatics became a science. Agenda was set by academics: software engineering primarily theoretical.
- European valleys: Utopian city planning, Sophia-Antipolis (1969)
- Computers in education: binary arithmetic, flow charts (Ecole, BASIC), programmed instruction, learning to think (and allowing teachers time and means to stimulate pupils to think)
- Appropriation: Teleac television courses, HCC and management (SSAA) - not politically motivated
- Squatter movement 1980s: inappropriating these computers when the HobbyCC were politically inspired and a lot of people among them.

Personal computers: more appropriation

Making the computer your own and creating new uses for computers.

Chaos Computer Club (1982): hacking as a way of making a political statement. War games (1983). Intergalactic foundation that hacked and tried to get their computerized versions of their worldview. Hacking was deeply political, in the early 80s it was something cool.

1984: Herwart Holland and Steffen Wernéry: hack Bildschirmtext (Btx), hacktivism.
1984: **Demoscenes**: mostly European phenomenon. Demo's on floppy disks built out of code and pictures; this was not the way the computer was intended. "diskmags".
1988: Wernéry arrested for computer crimes.

ALIEN and Blade Runner; shoot a commercial for a new computer in 1984, 'The Novel'.

Burgerinformatica (1983): Home computer projects were promoted by the government in the Netherlands.

- From 1986 onwards, it made computer education, as well as selling pc's, more profitable.
- The Netherlands: End 1980s: 30% owned a pc, by 1995 increased to 60%.
- Kees van Kooten, Wim de Bie (1986): ridiculed.

1993: Hacking at the end of the universe (a kind of festival for exchanging information)
1994: De Digitale Stad (funded for Amsterdam). A way to use computers to get together. Before anybody had thought about social media, it has been made by programmers in Amsterdam. Modems were sold out by February. When the Digital City went online, there was a meltdown because there was too much data that was transferred.

Rise of the Internet

Campbell-Kelly: “the confluence of three desires”.

France: Minitel (1978-2012) by France Telecom; only 8% of people only had a telephone connection. Very low number compared to rest of the world. Most small villages had a phone booth and that was perfectly fine. Minitel made sure that telephone connections became interesting to people, because they could get more than just a telephone connection. It was the newest tech that was coming right to your home. They gave these machines away (almost) for free. It looks like a teletext page, so it was less open than the internet and more confined. It kept France backwards.

Electronic mail: Companies, universities and government institutions. It needed explanation for people to understand what it was. It connected people.

The Netherlands:

- **Viditel** (1980): was more expensive
- Approximately simultaneously: **Teletext** (1980)
- **Datanet 1** (PTT)
  - Early fase:
    - circuit-switching on designated telex- lines
    - Successful with government circles, journalists, customs and banks
    - Use of own networks and interconnected designated lines (by PTT) was an alternative for large companies. PTT offered an “affordable” alternative.
  - Starting fase 1978-1981:
    - transition on packet switching on the basis of X25-protocol
  - Launched through existing network of telex-lines
  - Near the end:
    - 1948: boosted, GBA until end 1990s, then started using Internet (TCP/IP)
    - Until the end of 1990s for network payments
    - Until the end of 2000s in use with customs and *Rijksdienst voor het Wegverkeer*

Rise of game industry

Playing games is a way of **appropriating**:

- 1952: Chess algorithm by Alan Turing
- 1960: Nimbus
- 1970: Spacewar was created by MIT students in 1961 on a DEC PDP-1
  - To get families familiar with working with computers
  - Also, a reason to connect a monitor to your machine
- 1972: first (official) tournaments and Pong by Atari came out
- 1977: Game computers in arcade hall, Vectorbeam on a DEC PDP-architecture
- 1980: Pacman, by Namco
  - It appealed to many people
  - It was produced as software to buy in the store

- 1981: Donkey Kong by Nintendo
- 1991: Sonic the Hedgehog by Sega

This is the start of what the game world has become nowadays. It is now more commercial, more attention to gameplay, graphics and music. It takes wider forms, paying attention to serious games to learn people something.

Rise of computer science/information science

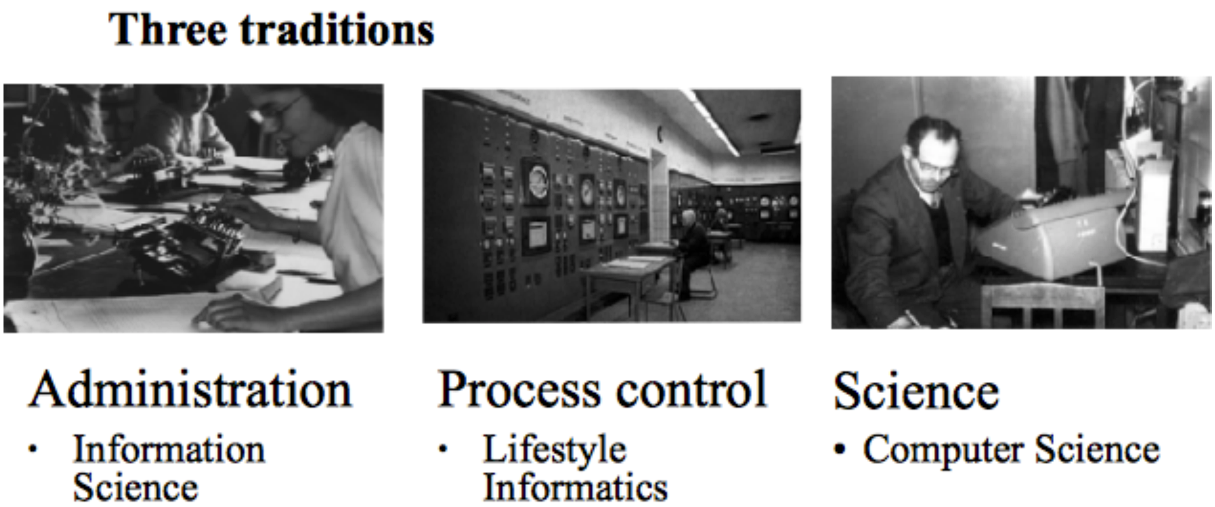
In France mainly from the departments of management (social) sciences or engineering schools. Since 1980s departments of computer science.

In The Netherlands, part of the departments of mathematics, sometimes in cooperation with department of physics or electronics. Since 1973: Graduation. 1981: First actual bachelor program. 1984: Bachelor in Information Science first established, at the Tilburg Catholic School.

In Belgium and United Kingdom information science is mostly part of the department of management sciences; lifestyle informatics would be part of the medicine department.

Step in the professionalization of the discipline. Other signs of professionalization:

- National and international organizations
- Journals (peer reviewed)
- Jargon
- Writing history (!)



HISTORY IS WRITTEN IN HINDSIGHT

If you look back in history and see what's there, it is quite common to see what you know today. It is difficult to get what you know today back in your head and look as history before itself on its own course. It has no effect at all, but it was there. Nobody knew at the time they were working on it that it would be here now.

Digital culture

Digital Culture = what has changed our outlook at the world. The way we look at the world nowadays.

- Globalization: also movies, television, telephone, etc.…
- Online library catalogues
- Desktop publishing
- Digital art (not necessarily digital, but culture)
    - New way of presenting art
- Online "meetings" (Derksen)
    - People were communicating in a different way
- **Gamification**: serious games

1980s: Information society

- Esther Dyson, *Release 2.1* (1998)
    - Communities around computers
        - The word community changed its meaning; no longer a group on one location
    - Content control
    - Communications Decency Act

Movies:

- 1990s: Knowledge society: computer graphics enhancing motion pictures
    - 1993: Jurassic Park
- 1993: Plan 9
- 1987: Minix
- 1999: The Matrix

End 1990s: Risk society

Risk society (information): about dealing with the risk and being aware that you have to deal with them. Politicized, certainly after 9/11. Rise of big data (thereby giving new birth to the information society) and new thoughts about privacy.

Information society, Knowledge society, Risk society Three ways of describing a view to digital culture

HC7 – Digital Culture/Divide

Recap

The **three traditions** that sparked the rise of computers:

- administration:
    - money, felt urgency to do things efficiently until WWII
- process control:
    - outlines were used but the analogue tech was more self-evident, 'computers' were used to make measurements and to react to those in an automated way. Automation self-evidently grew, it was a practice in itself

- science
    - calculations were more important

These 3 together are the history of our profession (CS, Information Science, AI). We can find the rise of automation as we know it today in these.

From distrust in machines to trust in **dinosaurs** in different traditions. Trust in machines that did not work: tubes or relais. They were selling them after WWII even if they weren't working very well. They bluffed their way in, but Babbage didn't do that. The US were far ahead with machines (more money, administrative tradition), they had a lot extra that EU did not, but we could make computers as well as they did. EU had more potential and the efficiency movement was still being funded, while in the US it was flourishing by itself. We had no companies producing machines capable of making many machines at once. Even if EU had some advantage in more tech, the US would easily bypass EU in producing machines.

Cold war: was a non-existing war; between SU and US and EU. Within US+EU there were tensions as well. US were overpowering in many aspects of life; American language became common language; the French were displeased about this. The EU wanted to confront the Defís American so they wanted their own machines; EU dwarves.it was a matter of national pride that every country had their own machine manufacturer. These would flourish until the '60s and they were there because they found a niche market. Some of the dwarves in US did that as well, but they needed that in EU to flourish.

Appropriating ACs; the word computer came later. Appropriating with the fact that they produced sounds (in the beginning). You could **hear** the machine working and that was the purpose; to see if it was working well. We use screens now, but they did not have that at the time.

Use of metaphor; the use of words like memory and stuff were used as metaphor but we started using those words now. The word computer (first meaning: person, next meaning: AC, last meaning: computer); *people familiarize themselves with the machine.* The computer was put out in very many types of fields. In US: psychologists, who were top notch computer scientists (learning how learning processes work, programming it on computer).

IBM sales tactic: FUD (fear, uncertainty and doubt) for not using ??. Big brother was a negative association. It kept IBM safe and big, also in EU where more than 70% of the market was mostly also American. **Real time computing**: things going wrong and losing money and losing faith; saved by the bell by succeeding. Why would you need real time? Waiting for a day is fine right? Nobody but military has use for this; IBM saved the day and made a lot of money. Civilian spin-off; ATM (NCR; US midget) and universal product code.

Agenda's: what did people want to achieve? Selling machines, academic discipline and thinking machines. IBM's agenda: selling machines and knowing what people needed. **Programming** and program library. **Unbundling** IBM (1968): big shift of the word '**software'**, it got its present meaning.

Software crisis; people were making software by themselves, there were no prices and it was sold anyway. The agenda was set by academics; and programming became a profession. It was no longer writing code and finding your way about the computer.

In EU there were no real **valleys**, but attempts in Twente and ?. Apart from these, the big companies moved into the valleys as well. What never succeeded was the informal and self-evidently growing of the valleys that was expected.

HCC; getting together to establish good practices of automation in Dutch companies. These were not politically motivated like in the US. All kinds of people from within several companies; we presented information as such, but never got an academic status. Not at the time. They did manage to get the message on national television; broadcasted many computer lessons, learning the Dutch audience to work with the computer, to program a bit, how a chip is designed. Except that most Dutch people did not have a computer yet.

Computers in education; binary arithmetic, flow charts, ECOL, BASIC, programmed instruction (trying to reduce the teaching load by having students check their own answers etc.) and learning to think. In several ways both computer and information science benefitted from each other.

1980s squatter movement: computer wizards and political motivation was coming together in one group. Within that group the ideas of modern computing were just as the 60s something you SHOULD learn about. You SHOULD be able to have access to internet.
1980s Demoscenes: mostly EU phenomenon, EU vs US in adoption of PC. Tech without consent, how to build your own modem, how to program your computer. New music, movies and a new cultural scene. One of these outcomes was *hacking at the end of the universe (Amsterdam, 1993)*; idea of digital city. You HAD to be there.

Rise of the Internet: “the confluence of three desires” (Campbell). The idea of having these global networks connected was in the US more self-evident than in EU. All of these local networks were natural. Things went different there. Connecting them was less self-evident, but it was easier to do, because you didn’t have that many networks. It slowed down the rise of internet. **Rise of game industry**.

Rise of computer science/ information science: In the Netherlands part of the departments of mathematics, sometimes in cooperation with department of physics or electronics. Graduation since 1973. First actual bachelor program dates 1981. Information science (SSAA): 1984 in Tilburg.

Digital culture: various people stimulated this rise. US: Esther Dyson (newsletter where people wanted to see all important people in CS and IS. People noted that changes were occurring and she made people aware that CS and AI is just as important as the machine itself. By producing these newsletters, she spurred the **information society** = she intended to indicate that having access to information was getting increasingly important in the world) Communities, content control, communications decency act.

- **Information society**: having access to information was getting increasingly important in the world

- **Knowledge society**: used by many groups who thought that info itself was not useful enough and that you should know how to handle the information
- **Risk society** (information): not about being aware of risks, but about dealing with the risk and being aware that you have to deal with them.

Why did we do this again?

Digital Culture

Ass1 question: *Describe the particular fact that you know that digital culture is at work. What are the pros and cons of that?* The culture has its historical roots. Understanding the rise of digital culture. Get out of comfort zone! Reflection on what we are, why we are (where we came from), where are we heading (and do we want to take another turn)?

Take away from this course: The simple story, people wanted to own a computer, so we started making them, just doesn't hold. It is about:

- Different desires
- People who just wanted to make money
- People with ideals
- Unintended use, appropriation
- Gaining trust
- Confluences
- Creating markets (creating desires!)
- Catching up with modernity (bluffing our way in!)

Digital culture: computer-like devices permeating every aspect of present-day society:

- Changing perception of society (information, knowledge, risk)
- Economical
- Cultural
- Social
- Entertainment
- Political
- Understanding / epistemological (the way we talk, the things we intend with words)

Digital culture: there is always a catch, always another side to the coin. That is NOT an advantage or disadvantage per se: it IS (so no reason to apologize). We get another outlook on life, on learning, on ourselves (do androids dream of electric sheep?). You can't always see it coming, but it's always there.
Library catalogues! Book to check the info on the net, or the net to check the info in a book.

Even simple stories can be complicated "everything can be found on the Internet": literally! Movie with person named Hal, which was intended as IBM (internet), but not true.

Special responsibility for IT professionals. Note that what is an advantage to one, may be a disadvantage to someone else. EXAMPLE: Health insurances get more and more expensive. Early 90s; start counting the seconds/minutes that every member of your staff attempts certain things he has to attend to (patients, administration, traveling etc.). Output: large database where you can count means! What was remarkable was that the deviations were abnormal. This made sense, not only because people do things differently, but also because some people will be sloppier in writing down the minutes than another. Sensible, but it changed the way Dutch healthcare could do its job. Not a bad thing, but people had to change the way their job worked.

1984 is getting closer than we want to and this is due to what we made possible. Changing the way we look at and perceive the world around us.

Digital divide

Digital divide = Belonging to digital culture. The pros of digital culture are not evenly divided. Two main definitions:

1. **Education** (age): difference in perception
2. **Geographical**: see the divide growing/changing in the course of years.

The Exam

1. Make sure your name and student number are on ALL your work. (2) Answer in English sentences. Spelling and grammar should be correct, either to the English or to the US standards. Just a few buzzwords never constitute an answer. Always explain yourself. Use appropriate examples to illustrate your answer.
2. This exam consists of 8 A-questions, 6 B-questions and 5 C- questions. The A-questions are about the lectures; the B-questions refer to information from the book by Campbell-Kelly. The C- questions require you to combine the information from the book and lectures in your answer.
3. You pick eight question: 3 A-questions, 3 B-questions and 2 C- questions. Each question is worth 1 point. Indicate clearly which questions you're answering by mentioning either the number and / or the title of the question in your answer. If you answer more than three A-questions, only the first three will be considered. Likewise, for the B- and C-questions.

Tell me a story and dazzle me with what you know. If your answer makes sense, you'll get a point.

Example Question (last year)

In the 1980s the personal computer became a more common tool in western households. Describe how this happened, and what made people think it might be worthwhile to have a pc in their house. Pay attention to the differences in perspective on both sides of the Atlantic.